

Firm-level price wedges:

how well does the mapping from portfolios to firms work?

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What this report checks

The paper estimates price wedges for characteristic-sorted decile portfolios and carries them to individual firms through a mapping from portfolio-level characteristics to portfolio-level wedges. This report asks how much that mapping can be trusted and whether it can be improved. Throughout, “pp” is percentage points of the price wedge, and a fitted or assigned wedge is what a mapping gives a portfolio or a firm from its characteristics.

Six checks, each answering a different question:

- Leave one characteristic out (Section 1): can the mapping price an anomaly it has not seen?
- Leave one family of characteristics out (Section 1): the same with a whole Freyberger–Neuhierl–Weber category held out, since value characteristics predict one another.
- Extreme deciles held out (Section 1): fit on deciles 2 to 9, predict 1 and 10; the mapping extrapolates at the firm level, where 94% of firm-months lie outside the range the portfolios span on at least one characteristic.
- Subsequent returns (Section 2): do firms called overpriced go on to underperform?
- Levels (Section 3): do firms realize the wedge they are assigned? This is the decisive test for a website that reports a number rather than a ranking.
- Stability across formation dates (Section 5): is the cross-section of wedges the same object in different parts of the sample?

Every check is scored on magnitudes as well as ordering: the realized wedge is regressed on the assigned one and the slope and intercept are reported. A perfect mapping has slope one and intercept zero. R^2 is computed out of sample as $1 - \text{SSE}/\text{SST}$ on the held-out portfolios, so a mapping that ranks well but mis-scales is penalized.

Which portfolios enter. Only characteristics whose construction is verified against the paper’s own data, as regressors and as portfolios; a portfolio we cannot reproduce carries a wedge we cannot trust. With 1 characteristic unverified (aPM), the rule costs nothing: the leave-one-family-out fit is 0.63 with every characteristic in and 0.62 without it.

1 The mapping on portfolios

Candidates: the paper’s three principal components of portfolio-level characteristics, ten principal components, and a penalized regression of the wedges on all 56 rank-normalized characteristics (penalty chosen out of sample, on the leave-one-family-out fit), with two variants of the latter,

weighting portfolios by the precision of their wedge and adding squared ranks. The paper fits on the 114 extreme deciles; here all 560 deciles are used, which is what the firm level needs.

Leave one characteristic out

Mapping	R^2	corr.	intercept	slope	RMSE
3 principal components (paper)	0.587	0.766	-0.19	0.959	6.23
10 principal components	0.747	0.864	-0.12	0.965	4.88
penalized regression	0.766	0.875	-0.11	0.959	4.70
penalized, precision-weighted	0.774	0.880	-0.35	0.944	4.63
penalized, with squared ranks	0.370	0.608	-1.39	0.626	8.48
squared ranks, precision-weighted	0.449	0.670	-1.31	0.640	8.09

Leave one family of characteristics out

Mapping	R^2	corr.	intercept	slope	RMSE
3 principal components (paper)	0.389	0.624	-1.68	0.665	8.15
10 principal components	0.467	0.683	-1.70	0.673	7.76
penalized regression	0.619	0.787	-1.01	0.843	6.14
penalized, precision-weighted	0.609	0.781	-1.30	0.805	6.33
penalized, with squared ranks	0.038	0.195	-4.89	0.047	39.54
squared ranks, precision-weighted	0.049	0.222	-4.88	0.056	37.44

Extreme deciles held out

Mapping	R^2	corr.	intercept	slope	RMSE
3 principal components (paper)	0.737	0.858	+1.02	0.963	7.23
10 principal components	0.805	0.897	+0.04	0.863	6.46
penalized regression	0.804	0.897	-1.35	0.853	6.56
penalized, precision-weighted	0.808	0.899	-1.37	0.856	6.50
penalized, with squared ranks	0.571	0.756	+0.11	0.798	9.51
squared ranks, precision-weighted	0.626	0.791	-1.52	0.699	9.72

The penalized regression is best on every check, by 0.2 of R^2 on the family test, and its slope is closest to one. Ten principal components come close; three do not. Neither variant helps: precision weighting has nothing to exploit (the standard errors of the portfolio wedges are nearly uniform, a 1.2-fold spread between the 10th and 90th percentiles, all near 29 pp because 463 formation dates that overlap in 179 of their 180 months are worth two or three independent observations), and squared ranks halve the out-of-sample fit. The penalty is the substance: without it the fitted wedges of an unfamiliar family are more than twice too dispersed (the realized wedge moves 0.42 per point of fitted wedge, against 0.89 with the penalty).

2 Subsequent returns

Firms are sorted into deciles on their assigned wedge each month and tracked forward; decile 1 is the most negative (most underpriced). If the wedge is mispricing that resolves, decile 1 should

outperform.

Mapping	months	weighting	decile 1	decile 10	spread
3 principal components	12	value	18.4	10.7	7.7
3 principal components	12	equal	26.3	11.5	14.8
3 principal components	36	value	60.3	34.8	25.5
3 principal components	36	equal	80.4	35.1	45.3
3 principal components	60	value	118.3	69.4	48.9
3 principal components	60	equal	143.2	67.0	76.2
10 principal components	12	value	21.3	10.5	10.7
10 principal components	12	equal	27.2	10.5	16.7
10 principal components	36	value	71.1	33.4	37.7
10 principal components	36	equal	84.1	31.6	52.6
10 principal components	60	value	134.3	65.6	68.8
10 principal components	60	equal	150.0	62.3	87.7
penalized regression	12	value	18.9	10.1	8.8
penalized regression	12	equal	25.5	10.0	15.5
penalized regression	36	value	64.6	31.9	32.7
penalized regression	36	equal	80.4	30.2	50.2
penalized regression	60	value	125.1	62.4	62.8
penalized regression	60	equal	145.9	58.9	87.0

Over five years the penalized regression separates the extreme deciles by 63 pp value-weighted and 87 pp equal-weighted, with deciles monotone except value-weighted at 36 months; three principal components give 49 and 76 pp and are not monotone value-weighted at 12, 36 and 60 months.

3 Levels: do firms realize the wedge they are assigned?

The portfolio-level tests cannot say whether the number the website prints for a firm is the right *level*, and two firm-level series can correlate at 0.95 while differing by 20 pp. This section tests levels directly (`realised_levels.py`). A candidate firm-level wedge is treated like a characteristic: firms are sorted into deciles on it every month at NYSE breakpoints, decile 1 holding the firms called most overpriced; each decile is held for fifteen years, equal-weighted and value-weighted; its realized price wedge is computed with the paper’s SDF and price of risk; and the realized wedge is regressed on the average wedge the candidate had assigned to the decile’s members. A candidate whose levels can be trusted gives a slope of one and an intercept of zero. The equal-weighted deciles ask whether firms realize what they are told, which is the website’s question; the value-weighted deciles ask whether the candidate respects the calibration. The calibration adds one more check: the value-weighted average of a candidate’s firm-level wedges must equal the market’s wedge, zero.

A reference point. The realized wedge of the equal-weighted market (one dollar in every firm) is -22.5 pp; that of the value-weighted market is 2.3 pp (zero by calibration, up to the sample filter). The average firm is a small firm and is far more underpriced than the average dollar. A firm-level wedge has to reproduce that gap.

In sample

Candidate	equal-weighted deciles			value-weighted		mean assigned	
	slope	icept	gap	slope	icept	EW	VW
Published firm-level wedges (paper)	0.64	-18	16.2	0.95	-8	-7	+9
Paper's construction, rebuilt	0.67	-20	18.0	1.02	-10	-6	+9
Value-weighted portfolios only (penalty 100)	0.67	-3	6.9	0.79	-3	-33	+1
Equal-weighted portfolios only (penalty 100)	1.05	+6	5.4	0.86	+5	-27	-8
Both portfolio sets (penalty 100)	0.86	-1	3.5	0.76	-4	-27	+1
Value-weighted portfolios only (penalty 300)	0.71	-3	5.9	0.85	-2	-31	+1
Equal-weighted portfolios only (penalty 300)	1.20	+9	6.8	1.09	+10	-27	-12
Both portfolio sets (penalty 300)	0.88	-1	4.5	0.84	-3	-27	+0

icept: intercept, pp. gap: mean absolute difference between assigned and realized across the ten deciles, pp. mean assigned: the average of the candidate's firm-level wedges, equal-weighted over firm-months (compare with -23) and value-weighted (compare with zero).

The paper's published firm-level wedges (three principal components of the 114 extreme deciles, each portfolio described by the equal-weighted average of its members' characteristics) have a slope of 0.64 and an intercept of -18 pp on equal-weighted deciles: the firms they call fairly priced realize about -25 pp, and only the most underpriced decile realizes what it is assigned. Their value-weighted average is +9 pp where the calibration requires zero. The reason is in Section 4.

The pipeline's mapping on value-weighted portfolios alone has the level right (intercepts -3 and -3 pp) and too much dispersion (slopes 0.67 and 0.79): it assigns the most underpriced decile about -70 pp where -42 is realized, because the typical firm sits far from the value-weighted portfolios' characteristics and the regression extrapolates. Fitting on both portfolio sets, each with the matching average of characteristics, does best: Both portfolio sets (penalty 100) gives a slope of 0.86 and an intercept of -1 pp equal-weighted, a mean absolute gap of 3.5 pp across deciles, 0.76 and -4 pp value-weighted, and a value-weighted average of +1 pp.

Decile detail

Assigned and realized wedge by decile of the assigned wedge, pp, for the published series and for both portfolio sets (penalty 100).

Published	1	2	3	4	5	6	7	8	9	10
equal-weighted, assigned	30	17	9	2	-4	-10	-15	-22	-29	-42
equal-weighted, realized	6	-2	-11	-28	-25	-29	-30	-30	-33	-40
value-weighted, assigned	31	17	9	2	-4	-10	-15	-21	-29	-39
value-weighted, realized	24	5	6	-7	-16	-26	-25	-25	-29	-47
Both portfolio sets (penalty 100)	1	2	3	4	5	6	7	8	9	10
equal-weighted, assigned	23	6	-3	-11	-18	-25	-31	-39	-48	-67
equal-weighted, realized	24	8	-3	-11	-16	-28	-33	-38	-43	-47
value-weighted, assigned	27	7	-3	-11	-18	-24	-31	-39	-48	-63
value-weighted, realized	27	3	-10	-19	-25	-28	-29	-31	-37	-45

Out of sample

Mappings fitted on formation dates before 1988 and evaluated on the dates from 1988 on, with the price of risk re-solved on each half so the market's wedge is zero in both (the published series, fitted on all dates, is evaluated on the same late dates).

Candidate	EW slope	EW icept	VW slope	VW icept
Published firm-level wedges (paper)	0.88	-27	1.16	-19
Value-weighted portfolios only (penalty 100)	0.72	-12	0.84	-13
Equal-weighted portfolios only (penalty 100)	1.03	-9	0.82	-7
Both portfolio sets (penalty 100)	0.87	-13	0.86	-12
Value-weighted portfolios only (penalty 300)	0.77	-13	0.90	-14
Equal-weighted portfolios only (penalty 300)	1.35	-1	1.20	-2
Both portfolio sets (penalty 300)	0.94	-11	0.97	-13

The slopes survive. The intercepts are about -12 pp for every candidate fitted on the early dates, because the late dates' average firm realized -32 pp against a late market of zero, below what the early dates taught the mapping. With the price of risk held at its full-sample value the intercepts are about $+15$ pp instead, the late market's own wedge. The ranking of firms carries forward; the level of the whole cross-section moves with the sample (Section 5).

4 Where the method weights by value and where by firm

Step	Paper	Pipeline
Market price of risk	value-weighted market wedge set to zero	same
Decile portfolios	value-weighted, NYSE break-points	same
Portfolio-level characteristics	equal-weighted average of the members' rank-normalized characteristics	capitalization-weighted average; both available
Regression across portfolios	each portfolio one observation, 114 extreme deciles	same, all 570 deciles
Firm-level regressions (investment, q)	each firm-month one observation	not run here

The calibration and the value-weighted decile define the object and stay. The problem is the average that describes a portfolio. A portfolio's equal-weighted average characteristics describe its typical member, a small firm; its wedge belongs to its dollars, which sit in its large members. Regressing the second on the first and evaluating at a firm's own characteristics hands every firm the big-firm level, which is the -18 pp intercept above. Pairing value-weighted wedges with capitalization-weighted averages is coherent but centres the mapping where almost no firm is; using both portfolio sets, each with its own average, is the construction that passes the level test and keeps the value-weighted average of firm-level wedges at zero. Rankings barely move under any of these; levels move by 10 to 20 pp for the typical firm. This is the evidence behind decision 1 in `decisions.pdf`.

5 Stability across formation dates

The price wedge of a portfolio is an average over formation dates, and so is the market price of risk. This section asks whether the cross-section of wedges is the same object in different parts of the sample. All 570 decile wedges are re-estimated on subsets of the formation dates, the market price of risk re-solved on each subset so that the market’s wedge is zero there (`stability.py`); the 560 portfolios of the verified characteristics are compared.

Formation dates	months	price of risk	corr. with all dates	slope	RMSE	sign changes
All dates, 1964–2011	559	3.53	1.00	1.00	0.0	0
Paper’s window, 1964–2002	462	3.97	0.99	1.06	1.4	1
First half, 1964–87	282	4.52	0.91	1.04	4.6	5
Second half, 1988–2011	277	2.86	0.76	0.83	6.9	8
1964–79	186	5.58	0.83	1.08	7.3	8
1980–95	192	2.49	0.17	0.10	10.9	29
1996–2011	181	2.85	0.76	1.23	10.2	5
From 1975	433	2.62	0.81	0.71	5.5	4
From 1980	373	2.64	0.76	0.70	6.3	7
From 1985	313	2.78	0.75	0.74	6.7	9
From 1990	253	2.90	0.77	0.93	7.4	5
From 1995	193	2.92	0.76	1.17	9.6	6
From 2000	133	2.16	0.52	0.77	12.2	11

corr., slope, RMSE: the subsample’s 560 wedges against those on all dates; sign changes: how many of the 56 long-short spreads change sign.

Two things stand out. Extending the paper’s window to all available dates changes almost nothing. Dropping the early dates does: every window starting in 1980 or later correlates at 0.5 to 0.8 with the full cross-section, and the 1980–95 third is nearly flat and unrelated to the rest.

The two halves. The halves use disjoint formation dates, so this is the cleanest comparison. Their wedges correlate at 0.45. Two halves of a constant cross-section would correlate at about 0.68, given the portfolio-specific estimation noise of each half (from a block bootstrap with the component common to all portfolios removed). The change across portfolios has a standard deviation of 11.0 pp, of which about 8.6 pp is beyond noise, on a cross-sectional spread of 9.3 pp. The cross-section moves between halves by roughly its own size. One caveat in the other direction: a block bootstrap with blocks shorter than the fifteen-year overlap understates noise, so part of the change may still be sampling; the gap between 0.45 and 0.68 is too large for all of it.

Where the change is. Long-short spreads (long side minus short side, pp) on all dates and in each half, for the characteristics that move most and least; 77% keep the sign of their spread across halves.

	all dates	1964–87	1988–2011
D2P	-16	-39	40
A2ME	-20	-42	22
E2P	-26	-51	12
ROC	-43	-61	-14
C2D	10	19	-15
SAT	-20	-3	-34
ROE	26	38	7
CAT	-14	3	-28
dPIA	-22	-23	-21
IDIOV	7	5	3
BEME	-48	-48	-47
R_36_13	-44	-43	-42
aSIZE	-22	-19	-20
Q	-45	-44	-44

The value ratios reverse between halves, profitability spreads shrink, momentum grows; book-to-market, q , size, long-term reversal and the investment sorts are the stable core.

At the firm level. The mapping refitted on each subsample and applied to the same firm-months, against the firm-level wedges from all dates (whose standard deviation across firms is 28 pp):

Fitted on	corr.	mean difference	s.d. of difference	firms moved > 10 pp
All dates, 1964–2011	1.00	+0.0	0.0	0%
Paper’s window, 1964–2002	1.00	+0.5	2.4	0%
First half, 1964–87	0.96	+4.8	7.5	27%
Second half, 1988–2011	0.92	+1.3	11.9	40%
1964–79	0.91	+12.1	12.1	63%
1980–95	0.44	+30.9	25.2	85%
1996–2011	0.90	-9.9	20.5	67%
From 1975	0.95	+5.1	9.2	35%
From 1980	0.91	+8.1	11.9	50%
From 1985	0.91	+4.6	11.5	42%
From 1990	0.92	-0.9	12.9	43%
From 1995	0.90	-7.4	18.8	63%
From 2000	0.75	+1.7	21.2	63%

Firm rankings are robust to the sample (correlations 0.90 to 0.96, the 1980–95 third and the post-2000 window aside); levels are not. The halves against each other at the firm level correlate at 0.79 with a standard deviation of the difference of 18 pp. This is the evidence behind decision 3 in [decisions.pdf](#): all available formation dates, the half-sample estimates as an internal check, no sample choices on the website.

6 Decisions taken, and what remains open

Three decisions follow from this report and are recorded, with their reasons, in [decisions.pdf](#): the firm-level wedge is the equal-weighted realized wedge of firms with the firm’s characteristics with the value-weighted average pinned at zero (Section 3); the characteristics are defined as their

source papers define them (`signals.pdf`); the wedges use all available formation dates, with no sample choices on the website (Section 5). Every build reports the level test, the aggregate check, the family-out fit, the subsequent-return spreads and the half-sample comparison before a series is published.

Open: sensitivity to the fifteen-year horizon and to the SDF is untested; the firm-level regressions in the paper that use the wedge as a regressor in levels inherit the intercept documented in Section 3 and would change with a level-calibrated series.